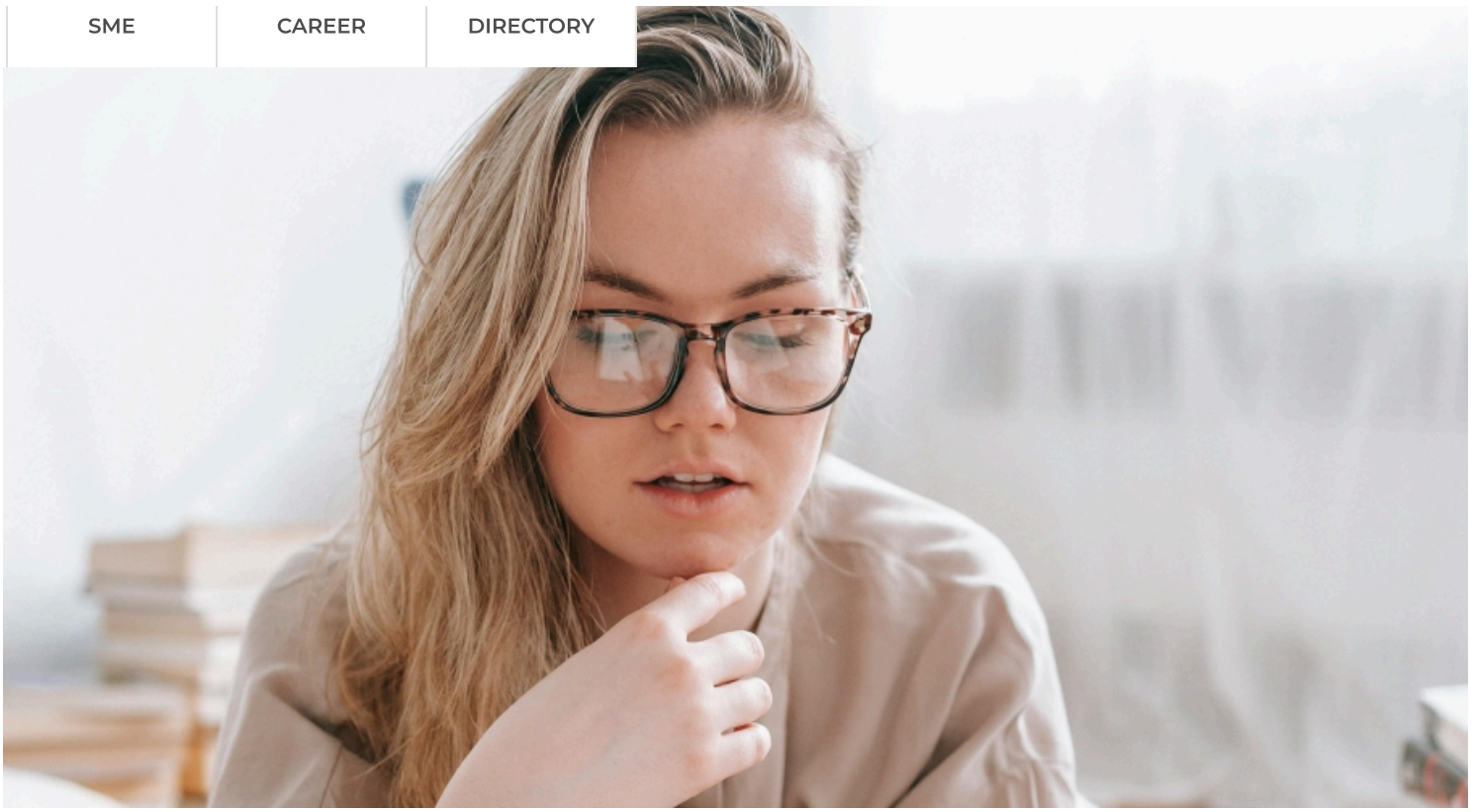

New research reveals how organisations can identify potential leaders based on cognitive and behavioural signals instead of relying on formal experience.

For decades, leadership identification has relied on personality inventories, interviews, and performance history. These tools are valuable, but they often capture who people are, not how they think and adapt under pressure. Research from Elizabeth "Zab" Johnson and Michael Platt of the Wharton Neuroscience Initiative (WiN), Korn Ferry, and Lazul.ai, shows how neuroscience-informed, AI-enabled assessments can add a powerful new layer to leadership pipelines, especially at early career stages.

What the research found

By studying undergraduate students well before formal management experience, the research team identified early cognitive and behavioural signals associated with leadership propensity. The approach combined:

- Psychometrics from Korn Ferry, a global organisational consulting firm known for leadership and talent assessment.
- Neuroscience-based, game-driven assessments from Lazul.ai, a platform that uses cognitive science to measure how people think, adapt, and manage uncertainty in real time.
- Wharton Neuroscience's research expertise, integrating behavioural science, decision-making, and analytics to model leadership emergence.

The results point to a more nuanced view of leadership potential, one that goes beyond charisma or résumé signals.

Key takeaways for people analytics leaders

- Leadership is visible in decision-making patterns, not just traits. Students who demonstrated cognitive flexibility (quickly switching strategies under pressure) were more likely to take on leadership roles.
- How people allocate attention matters. Future leaders were more likely to distribute effort across multiple priorities rather than focus narrowly on a single high-reward task.
- Risk tolerance is multidimensional. Comfort with uncertainty emerged from how individuals reasoned through trade-offs, not from assertiveness or outgoing personality alone.

Importantly, these signals appeared before individuals entered the workforce, suggesting leadership potential can be identified much earlier than traditional models assume.

Why this matters for CHROs and talent leaders

methods.

surface high-potential individuals who may not fit conventional leadership molds.

- Build broader, more diverse pipelines. By focusing on cognitive and decision-making processes rather than polished leadership behaviours, organisations may reduce bias and expand who gets recognised as "leadership material."

This article was partially generated by AI and edited (with additional writing) by Kyle Kearns.

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